

Aru

A Local Epistemic Integrity Engine for Non-Directive Civic Reflection

Working White Paper - Build Draft v0.1

Status: Foundational redesign draft

Supersedes: Earlier "community mirror" framing

Primary audience: First serious reviewers, technical collaborators, governance collaborators, privacy/data advisers, and institutional design reviewers

Prepared: 24 August 2026

Contents

Version Note

Abstract

Part I - The Core Frame

1. Purpose of This Paper
2. Core Thesis
3. From Community Mirror to Epistemic Integrity
4. System Identity: Zone Trip, Aru, and the Broader Class

Part II - The Problem

5. The Civic Problem
6. The Epistemic Problem
7. Reference Case: Burning Man and Cultural Drift
8. The Necessary Self-Challenge

Part III - Epistemic Object Model

9. Why Object Status Matters
10. Primary Object Types
11. The Rule Against Ontological Collapse
12. Epistemic Debt

Part IV - Transformation Warrants

13. What Is a Transformation Warrant?
14. Core Transformation Rules

Part V - Contextual Competence

15. The Need for Context
16. Core Context Doctrine
17. Types of Context
18. Prohibition on Community-Belief Priors

Part VI - Locality, Burn, and Auditability

19. Local Containment
20. Nightly Data Burn
21. Auditability Without Reconstruction

Part VII - Aggregation as Epistemic Transformation Pipeline

22. Pipeline Overview
23. Layer Functions

Part VIII - Ethical Charter

24. Purpose of the Ethical Charter
25. Charter Principles

Part IX - Constitutional Requirements

26. Purpose of the Constitution

27. Constitutional Articles Required

Part X - Separation of Powers

28. Why Separation of Powers Is Necessary

29. Powers to Separate

Part XI - Constitutional Interpretation

30. Interpretation Rules

Part XII - Failure Modes

31. Primary Failure Modes

Part XIII - Deployment Model

32. Deployment Conditions

33. Prototype Priorities

Part XIV - Reviewer Objections

34. Objection: This Still Launders AI Authority

35. Objection: Auditability Conflicts With Deletion

36. Objection: Regional Context Will Become Stereotyping

37. Objection: A Non-Directive System Still Influences Behaviour

38. Objection: This Is Overbuilt

Part XV - Working One-Page Summary

39. Aru in Brief

Part XVI - Current Build Questions

40. Current Build Questions

Version Note

This paper supersedes the earlier Zone Trip white paper that framed the project primarily as a "community mirror." That metaphor remains useful for public explanation, but it is no longer sufficient as the core technical or constitutional description.

The stronger frame is that **Aru is a local epistemic integrity engine.**

Aru does not simply reflect a community back to itself. It receives anonymous participant expression and transforms it through a governed sequence of redaction, observation, interpretation, clustering, synthesis, constitutional review, and public reflection. Each transformation creates risk. Meaning may be overstated. Interpretation may be mistaken for fact. Repetition may be mistaken for importance. Synthesis may be mistaken for consensus. Context may be mistaken for evidence. AI fluency may be mistaken for understanding.

This paper therefore treats Aru as a system whose primary function is not to maximise insight, but to preserve the integrity of the path from participant utterance to public reflection.

Abstract

Aru is a local, non-directive epistemic integrity engine for anonymous civic reflection.

It is designed for contexts in which a community may benefit from encountering patterns, tensions, contradictions, absences, and questions emerging from participant expression, without converting that expression into governance, surveillance, polling, behavioural direction, institutional intelligence, or false consensus.

Aru's core purpose is not to discover "what a community thinks." It is to constrain what may responsibly be said from the material participants offered.

The system is built around five central commitments.

First, **epistemic object separation**. Participant utterances, extracted observations, candidate interpretations, clusters, themes, syntheses, and public reflections are different object types. They must not collapse into one another.

Second, **provenance and epistemic debt**. Every derived object must carry the burden of where it came from, what transformation produced it, what uncertainty attaches to it, and what it is forbidden to claim.

Third, **non-direction**. Aru may surface tensions, contradictions, and questions. It must not produce recommendations, rankings, mandates, action plans, behavioural nudges, or institutional advice.

Fourth, **local containment and nightly burn**. Raw participant material should be processed locally and destroyed through a nightly data burn. The system may preserve only non-identifying epistemic provenance and constitutionally approved derived material necessary for audit and public reflection.

Fifth, **constitutional governance of interpretive authority**. Aru's constitution must govern not only data protection and misuse, but the authority of interpretation itself.

The central claim of this paper is:

Aru exists to preserve epistemic integrity across the transformation of anonymous participant expression into public reflection, so that no stage of interpretation silently gains more authority than the evidence can support.

Part I - The Core Frame

1. Purpose of This Paper

This paper is written as a build document for serious scrutiny.

It is not a pitch. It is not a manifesto. It is not a promotional explanation of an AI installation. It is a structured account of what Aru is, what problem it addresses, what powers it must not acquire, and what architectural constraints are required for it to remain legitimate.

The paper has four purposes.

First, to define the new frame: Aru as an epistemic integrity engine rather than merely a community mirror.

Second, to distinguish the project container, the system, and the broader architectural class.

Third, to describe the epistemic object model and transformation pipeline at the centre of the system.

Fourth, to establish the constitutional and ethical requirements that must govern any prototype or deployment.

The paper should be read with a hostile but fair question in mind:

Does this system genuinely preserve epistemic integrity, or does it create a more sophisticated way to launder authority through AI-generated reflection?

That is the correct scrutiny standard.

2. Core Thesis

Aru's core thesis is:

Anonymous civic reflection can be useful only if the system transforming participant expression into public output preserves the epistemic status, provenance, uncertainty, and limits of every derived claim.

This means Aru is not designed to maximise output quality in the ordinary AI sense. It is not trying to produce the most compelling summary, the most emotionally resonant synthesis, the most actionable insight, or the clearest diagnosis of a community.

Aru is designed to prevent overclaiming.

It should make visible only what the participant material can responsibly support. It should preserve ambiguity where ambiguity remains. It should distinguish observation from interpretation. It should distinguish recurring language from representative sentiment. It should distinguish contextual understanding from evidence. It should distinguish public reflection from mandate.

Aru's purpose is not:

- to discover what the community thinks;
- to identify the most important issues;
- to rank concerns;
- to advise organisers;
- to diagnose cultural health;
- to predict drift;
- to produce institutional intelligence;
- to measure sentiment;
- to generate recommendations.

Aru's purpose is:

- to receive anonymous participant expression;
- to process it through constrained local interpretation;

- to preserve object status and provenance across transformations;
- to identify tensions, contradictions, absences, and patterns without overclaiming;
- to produce a constitutionally reviewed public reflection;
- to leave agency with the community.

The practical design doctrine is:

Aru does not discover what a community thinks. Aru constrains what may responsibly be said from the material participants offered.

3. From Community Mirror to Epistemic Integrity

The earlier framing described Zone Trip/Aru as a "community mirror." That frame remains useful, but it is incomplete.

A mirror suggests direct reflection. Aru is not direct. It transforms.

Participant speech does not move unchanged into public reflection. It passes through capture, transcription, redaction, eligibility checks, semantic parsing, interpretation, clustering, theme formation, synthesis, constitutional filtering, human review, and publication. Every stage creates a new object. Every new object has a different relationship to evidence.

This creates a central risk: **epistemic inflation**.

Epistemic inflation occurs when a derived object gains more authority than its source justifies.

Examples:

A participant utterance becomes treated as a reliable observation.

An observation becomes treated as an interpretation.

An interpretation becomes treated as fact.

A cluster becomes treated as a theme.

A repeated theme becomes treated as importance.

A synthesis becomes treated as consensus.

A public reflection becomes treated as what "the community thinks."

Aru exists to prevent these authority jumps.

The mirror metaphor can still describe the participant-facing experience. A person may enter a booth, speak, and later encounter a public reflection of material offered by participants. But internally, Aru should not be understood as a mirror. It should be understood as a governed epistemic transformation system.

The question is not simply:

Can the community see itself?

The better question is:

Can a system transform participant expression into public reflection without overstating what the material justifies saying?

4. System Identity: Zone Trip, Aru, and the Broader Class

This paper separates three levels of identity.

4.1 Zone Trip

Zone Trip is the originating project and deployment container. It includes the cultural frame, the physical installation, the research lineage, the civic-artistic setting, and the broader question of whether a community can encounter itself without being directed.

Zone Trip is the world in which the first version of this work is likely to appear.

4.2 Aru

Aru is the governed interpretive engine inside the project.

Aru is responsible for receiving anonymous participant expression, processing it locally, preserving epistemic status across transformations, enforcing constitutional limits, and producing non-directive public reflection.

Aru is not identical to the booth. It is not identical to Zone Trip. It is the epistemic system.

4.3 Epistemic Integrity Engine

An epistemic integrity engine is the broader architectural class that Aru may help define.

Future systems may use related methods for other domains: public sensing, local awareness, claim provenance, institutional briefings, journalist tools, or civic knowledge systems. Those possibilities should not be collapsed into Aru's first deployment.

Aru 1 remains a non-directive anonymous civic reflection system. Later systems may inherit some methods, but they would require separate constitutions, purposes, and constraints.

The distinction matters because premature generalisation would endanger the original project. Aru should not become a broad civic intelligence platform by accident.

Part II - The Problem

5. The Civic Problem

Communities often struggle to see themselves clearly.

This is not because people are unintelligent. It is because communal self-understanding is difficult under conditions of scale, symbolic complexity, institutional pressure, cultural drift, social fragmentation, and unequal voice.

A community may contain:

- founders and newcomers;
- builders and consumers;
- organisers and critics;
- institutional stewards and informal cultural carriers;
- people with memory and people with fresh perception;
- people who feel protective and people who feel excluded;

dominant stories and unsaid tensions.

Ordinary community reflection happens through conversation, ritual, conflict, gossip, art, shared labour, public meetings, private frustration, humour, and silence. These forms are valuable because they are human. But they may not travel evenly across a large or complex community.

Aru is proposed for the narrow case where reflection is dispersed and difficult to hold collectively.

It should not be assumed that every community needs such a system. Some communities would be better served by direct conversation, leadership, accountability, rest, ritual, or silence. Some would be harmed by an AI-mediated reflective system. Some would misuse it. Some would treat it as a mandate regardless of design.

Aru must therefore justify each deployment.

6. The Epistemic Problem

The deeper problem is not simply that communities lack reflection. The deeper problem is that systems of reflection can become systems of false authority.

When participant material is processed into a public output, several corruptions are possible.

A system can convert partial material into representative claims.

It can convert repeated language into apparent importance.

It can convert AI-generated fluency into apparent understanding.

It can convert interpretation into fact.

It can convert synthesis into consensus.

It can convert context into evidence.

It can convert public reflection into institutional legitimacy.

These are epistemic failures before they are political failures.

They concern what the system is allowed to know, infer, claim, preserve, and publish.

Aru therefore requires an architecture that controls not only data flows, but also the authority of derived meaning.

7. Reference Case: Burning Man and Cultural Drift

Burning Man remains a useful reference case because it shows how living communities can change while preserving their original language.

The relevant issue is not whether Burning Man has failed. That framing is too crude. The relevant issue is how cultural systems behave under scale, prestige, institutionalisation, money, myth, participation, and memory.

Burning Man offers visible tensions that are useful for Aru's design enquiry:

- participation versus consumption;
- gifting versus prestige;
- decommodification versus money;
- memory versus renewal;
- founder myth versus institutional continuity;
- radical self-reliance versus accessibility;
- informal culture versus formal governance;

principles as living practice versus principles as brand language.

The value of the Burning Man case is not that Aru is built to analyse Burning Man. It is that Burning Man makes the general problem legible.

When a community becomes large, symbolically powerful, and institutionally necessary, it may require forms of reflection. But any instrument of reflection can become an instrument of authority.

That is the problem Aru is designed to confront.

8. The Necessary Self-Challenge

Aru must challenge its own necessity before every deployment.

The existence of cultural drift does not automatically justify an AI-mediated reflection system. A community may need better human practices, not a machine. It may need direct accountability, not synthesis. It may need stronger rituals, not analysis. It may need courageous leadership, not anonymous input. It may need to tolerate conflict rather than process it.

Aru should not deploy where it would replace organic reflection, launder institutional authority, increase dependence on mediated synthesis, or create a new centre of interpretive power.

Before deployment, the project must ask:

Does this community need a reflection system, or are we imposing one?

Will Aru make the community more capable of speaking to itself, or more dependent on machine synthesis?

Will Aru clarify uncertainty, or convert uncertainty into elegant language?

Will participants understand what the system is and is not?

Will the host treat the output as reflection, or as proof of having listened?

Will the public output be used as a mandate?

Can the system remain local?

Can the nightly burn be enforced?

Can epistemic provenance survive without preserving reconstructive participant data?

Can regional context improve interpretation without becoming prior-driven judgement?

If these questions cannot be answered within Aru's constitutional purpose, deployment should be refused.

Part III - Epistemic Object Model

9. Why Object Status Matters

Aru's most important architectural principle is epistemic object separation.

The system must treat each stage of transformation as producing a different kind of object. These objects must not silently collapse into one another.

A participant utterance is not the same thing as an extracted observation.

An extracted observation is not the same thing as an interpretation.

An interpretation is not the same thing as a theme.

A theme is not the same thing as a community belief.

A synthesis is not the same thing as consensus.

A public reflection is not the same thing as truth.

If these distinctions are not preserved, Aru becomes unsafe even if it remains private and local.

Privacy alone is not enough. A perfectly private system can still overclaim. A local system can still misrepresent. A nightly burn can still leave behind distorted derived objects. A non-surveillance system can still produce false authority.

Aru must therefore govern meaning status, not only data retention.

10. Primary Object Types

10.1 Participant Utterance

A participant utterance is the original expression offered by a participant.

It may be spoken or typed. It may be coherent, fragmented, emotional, ambiguous, contradictory, symbolic, humorous, accusatory, confused, or precise.

Its authority is limited.

A participant utterance is evidence that a participant said something. It is not automatically evidence that the statement is true, representative, important, or shared.

Aru must not treat utterance as fact.

10.2 Redacted Utterance

A redacted utterance is a version of the participant utterance from which identifying or unsafe material has been removed or masked.

Its purpose is to reduce exposure while preserving enough meaning for interpretation.

Its authority remains limited. Redaction may remove context. Therefore a redacted utterance may be less complete than the original and should carry that limitation.

10.3 Extracted Observation

An extracted observation is a system-generated representation of something that appears in the utterance.

Example:

Participant utterance: "It feels like people arrive now expecting the city to be ready for them."

Possible extracted observation: "The participant describes concern about arrival, readiness, and expectation of provision."

This is not yet interpretation. It is an attempt to state what appears in the utterance in a more structured form. An extracted observation must be recoverable from the utterance. If it cannot be traced back to the offered material, it should not exist.

10.4 Candidate Interpretation

A candidate interpretation is a possible meaning inferred from an observation.

Example:

Observation: "Concern about arrival, readiness, and expectation of provision."

Candidate interpretation: "This may relate to a tension between participation and consumption."

The word "may" matters. A candidate interpretation is not a finding. It is a plausible reading.

Aru should preserve alternative interpretations where relevant.

10.5 Cluster

A cluster is a grouping of related observations or candidate interpretations.

Clusters may form around shared language, related tensions, symbolic motifs, repeated concerns, or structural similarity.

A cluster is not a vote. It is not a faction. It is not proof of prevalence. It is not automatically important because it contains multiple items.

A cluster's authority is relational: it says that certain objects appear meaningfully related under a stated grouping logic.

10.6 Theme or Tension

A theme or tension is a higher-level object inferred from clusters or recurring interpretive structures.

A theme may describe a field of concern. A tension may describe an unresolved pressure between competing values or experiences.

Example:

"A tension appears around participation and provision."

This must not become:

"The community wants less provision."

Themes and tensions must carry uncertainty, scope limits, and non-representative framing.

10.7 Synthesis

A synthesis is a composed account of multiple themes, tensions, contradictions, absences, or symbolic patterns.

Synthesis is especially dangerous because it can sound authoritative. The more coherent the synthesis, the greater the risk that readers treat it as settled knowledge.

A synthesis must therefore remain visibly bounded.

It should state what appeared in the material, not what the community is.

10.8 Public Reflection

A public reflection is the constitutionally reviewed output offered back to the community.

It is the most visible object and therefore the most dangerous.

It must not claim to represent the whole community.

It must not instruct.

It must not rank.

It must not diagnose.

It must not recommend.

It must not identify factions.

It must not imply consensus.

It must not hide uncertainty.

A public reflection is not a conclusion. It is a constrained civic artifact.

11. The Rule Against Ontological Collapse

Aru must prevent ontological collapse.

Ontological collapse occurs when one object type is treated as another.

Examples:

A participant utterance is treated as fact.

A cluster is treated as a faction.

A repeated concern is treated as community priority.

A theme is treated as consensus.

A synthesis is treated as diagnosis.

A public reflection is treated as mandate.

This is one of the central constitutional risks.

Aru's architecture must mark and preserve object type throughout the pipeline. Every object should carry metadata or an equivalent internal record identifying:

- object type;
- source dependency;
- transformation stage;
- confidence or uncertainty status;
- prohibited claims;
- retention status;

audit status.

This does not mean preserving raw participant content indefinitely. It means preserving the epistemic status of derived material.

12. Epistemic Debt

Every derived object must carry its epistemic debt.

Epistemic debt means the object remains accountable to:

- what it came from;
- what transformation produced it;
- what uncertainty attaches to it;
- what has been removed or redacted;
- what alternative interpretations remain plausible;

what it is forbidden to claim.

A derived object that cannot state its debt should not be used in public reflection.

This is central to Aru's integrity. The system should never allow a clean sentence to hide a fragile chain of inference.

For example:

"A tension appears around participation and provision."

This sentence carries epistemic debt. It depends on participant material, redaction, observation extraction, candidate interpretation, clustering, and synthesis. It may be valid as a bounded reflective statement. It is not valid as a claim that the community prioritises participation over provision.

The public may see only the final reflection, but the system must retain enough non-identifying epistemic provenance to audit whether the sentence was responsibly produced.

Part IV - Transformation Warrants

13. What Is a Transformation Warrant?

A transformation warrant is the reason Aru is allowed to move from one object type to another.

It answers:

Why is this transformation permitted?

What evidence supports it?

What uncertainty remains?

What authority does the new object have?

What authority is it prohibited from claiming?

The transformation warrant is the system's defence against silent authority gain.

Without warrants, the pipeline becomes ordinary AI summarisation. With warrants, each movement from utterance to reflection must justify itself.

14. Core Transformation Rules

14.1 Utterance to Redacted Utterance

Permitted when redaction is necessary to remove identifying, unsafe, or constitutionally prohibited material.

The transformation must preserve meaning where possible, but participant protection overrides semantic completeness.

Risk: over-redaction may remove context; under-redaction may expose participants.

Required status: redacted, meaning-limited, non-identifying.

14.2 Redacted Utterance to Extracted Observation

Permitted when the observation is directly recoverable from the redacted utterance.

The observation must not add interpretation disguised as description.

Risk: the system may smuggle interpretation into observation.

Required status: descriptive, source-dependent, non-final.

14.3 Observation to Candidate Interpretation

Permitted when the interpretation is plausible and marked as interpretive.

The system should preserve alternative interpretations where relevant.

Risk: plausible interpretation becomes treated as fact.

Required status: candidate, uncertain, non-exclusive.

14.4 Candidate Interpretation to Cluster

Permitted when multiple objects show meaningful relation under a defined grouping logic.

Grouping may be semantic, symbolic, structural, or tension-based. It must not be merely convenient.

Risk: cluster becomes treated as prevalence, faction, or importance.

Required status: relational, non-representative, non-metric.

14.5 Cluster to Theme or Tension

Permitted when a coherent higher-level pattern can be stated without overclaiming.

The theme or tension must not imply majority support, consensus, or priority.

Risk: pattern becomes treated as community belief.

Required status: bounded, reflective, non-representative.

14.6 Theme or Tension to Synthesis

Permitted when multiple themes or tensions can be composed without erasing contradiction.

The synthesis must preserve uncertainty and avoid closure.

Risk: synthesis becomes diagnosis.

Required status: interpretive, limited, contradiction-preserving.

14.7 Synthesis to Public Reflection

Permitted only after constitutional review.

The public reflection must comply with non-direction, anti-metric capture, anti-surveillance, anti-factionalisation, non-representativeness, and participant protection.

Risk: reflection becomes mandate or authority object.

Required status: public, bounded, non-directive, non-representative.

Part V - Contextual Competence

15. The Need for Context

Aru may need contextual competence to interpret participants faithfully.

Participants may use local idioms, place names, cultural references, camp names, institutions, event-specific language, humour, slang, historical references, or terms that are meaningful only within a particular community.

Without context, Aru may misread them.

A local phrase may be interpreted literally when it is idiomatic.

A reference to an institution may be missed.

A joke may be treated as a claim.

A ritual term may be treated as ordinary language.

A local grievance may be misclassified.

A speech pattern may be mistaken for incoherence.

Therefore Aru may require a governed contextual layer.

But this layer is dangerous.

16. Core Context Doctrine

The governing rule is:

Context may resolve meaning; it may not supply position.

This means regional or deployment-specific context may help Aru understand what a participant is referring to. It may not determine what the participant believes, feels, intends, or represents.

Permitted:

Aru uses local context to understand that a phrase refers to a specific place, practice, institution, ritual, camp, or event.

Prohibited:

Aru uses local context to infer that because people in this community "usually mean X," this participant probably means X.

Context can expand the interpretations Aru is capable of recognising. It cannot decide which interpretation the present evidence supports.

The present participant utterance must remain the controlling evidence.

17. Types of Context

Aru should distinguish at least four context sources.

17.1 Base Model Competence

General language competence, general cultural knowledge, and ordinary reasoning already present in the model.

Risk: hidden assumptions.

17.2 Governed Regional Context

Pre-approved contextual information about local language, geography, institutions, event terms, cultural references, and deployment-specific vocabulary.

Risk: context becomes stereotype or prior.

17.3 Validated Deployment-Learned Context

Contextual knowledge learned during deployment that improves interpretation of references or language without forming beliefs about the community.

Example: a repeated unfamiliar term is validated as the name of a specific installation.

Risk: the system starts learning community-belief priors.

17.4 Current Participant Evidence

The actual utterance or contribution being processed.

This must have the highest authority for interpreting that participant.

The rule is simple:

Prior context may expand what Aru can understand. It must not overpower what the participant actually said.

18. Prohibition on Community-Belief Priors

Aru should be prohibited from retaining or applying community-belief priors.

Community-belief priors include assumptions such as:

"People here usually believe X."

"Veterans tend to mean Y."

"Newcomers are usually concerned about Z."

"This faction normally complains about A."

"This region is distrustful of B."

"This type of participant usually supports C."

These priors are dangerous because they create a behavioural model of the community. They place interpretive pressure on future participants. They turn historical pattern into hidden evidence.

Aru may retain interpretive competence. It must not retain assumptions about what future participants are likely to mean or believe.

Permitted:

"This phrase is commonly used locally to refer to a particular road."

Prohibited:

"Participants who mention this road are usually criticising organisers."

The first resolves reference. The second supplies position.

Part VI - Locality, Burn, and Auditability

19. Local Containment

Aru should operate locally by default.

Raw participant material should not leave the deployment environment. The system should not rely on continuous internet connectivity for capture, transcription, redaction, semantic parsing, aggregation, or constitutional filtering.

This is not merely a privacy preference. It is part of the epistemic architecture.

A participant should be able to understand the boundary:

Their contribution is processed inside the local system.

It is not sent to external AI providers.

It is not stored in a cloud account.

It is not used to train general systems.

It is not available for later institutional analysis.

Raw material is burned nightly.

A more powerful connected system may be rejected if it weakens participant trust or deletion integrity.

Aru's goal is not maximum intelligence. It is bounded, auditable, constitutionally safe interpretation.

20. Nightly Data Burn

Aru should perform a nightly data burn at the end of each deployment day.

The burn should destroy:

- raw audio;
- raw transcripts;
- temporary transcription files;
- redaction working files;
- prompt histories containing participant material;
- processing caches;
- embeddings linked to individual utterances;
- session-level files capable of reconstruction;
- diagnostic logs containing participant material;

any intermediate material that could reasonably reconstruct what a participant said.

The nightly burn prevents raw participant expression from becoming a research archive, surveillance dataset, institutional memory bank, legal exposure, or future training corpus.

The system may listen, but it must not keep.

21. Auditability Without Reconstruction

The nightly burn creates a hard design tension.

Aru must destroy raw and reconstructive material. But it must also preserve enough non-identifying epistemic provenance to audit whether the public reflection was responsibly produced.

This means the system must distinguish between two kinds of provenance.

21.1 Content Provenance

Content provenance preserves enough material to reconstruct what participants said.

This is dangerous and should generally burn.

21.2 Epistemic Provenance

Epistemic provenance preserves the status of transformations without preserving reconstructive participant content.

This may include:

- object type;
- transformation stage;
- whether redaction occurred;
- whether the object was eligible for aggregation;
- uncertainty status;
- whether alternatives were preserved;
- whether contextual information was used;
- whether contextual information resolved meaning or supplied position;
- whether the output passed constitutional review;
- whether the source material was burned;

deletion verification records.

The unresolved design problem is how much epistemic provenance can survive without becoming reconstructive.

This paper does not pretend that question is solved. It identifies it as a primary engineering and governance problem.

Part VII - Aggregation as Epistemic Transformation Pipeline

22. Pipeline Overview

The aggregation layer should be redesigned as an epistemic transformation pipeline.

Its sequence is:

- Participant utterance
- > Redacted utterance
- > Extracted observation
- > Candidate interpretation
- > Cluster
- > Theme or tension
- > Draft synthesis
- > Constitutional public reflection

The pipeline should not be evaluated by how compelling the final reflection sounds. It should be evaluated by whether each transition preserved object status, uncertainty, provenance, and constitutional limits.

23. Layer Functions

23.1 Capture Layer

Receives participant expression without identity capture.

Must not require login, phone pairing, ticket scan, demographic profile, device identifier, or account creation.

23.2 Redaction Layer

Removes or masks identifying, unsafe, or constitutionally prohibited material.

Must operate as early as possible.

23.3 Eligibility Layer

Determines whether material can enter aggregation, requires redaction, must be quarantined, or must be destroyed.

Exclusion should be constitutional, not editorial.

Material should not be excluded merely because it is critical, uncomfortable, strange, emotional, unpopular, or inconvenient.

23.4 Observation Extraction Layer

Converts redacted utterances into recoverable observations.

Must not smuggle interpretation into description.

23.5 Candidate Interpretation Layer

Generates possible meanings while marking them as interpretive and uncertain.

Should preserve alternatives where relevant.

23.6 Clustering Layer

Groups related observations or interpretations.

Must not convert grouping into prevalence, priority, consensus, or faction.

23.7 Theme and Tension Layer

Forms bounded themes, tensions, contradictions, absences, and symbolic patterns.

Must not produce rankings, counts, or claims of representativeness.

23.8 Synthesis Layer

Composes public-facing draft material.

Must preserve uncertainty and contradiction.

Must avoid conclusion, diagnosis, advice, and institutional usefulness.

23.9 Constitutional Output Filter

Checks whether the draft violates non-direction, anti-metric capture, anti-surveillance, anti-factionalisation, non-representativeness, local containment, burn requirements, or object-status discipline.

23.10 Human Steward Review

Reviews the draft for constitutional compliance.

Stewards do not make the output more institutionally acceptable. They protect the integrity of the system.

23.11 Public Reflection Layer

Publishes a bounded reflection.

Must include a clear statement that the output is reflective, not representative; interpretive, not directive; and not a mandate.

Part VIII - Ethical Charter

24. Purpose of the Ethical Charter

The ethical charter defines the commitments that govern Aru before technical implementation, deployment, funding, or institutional use.

The charter is not a values statement added to a system. It is part of the system's design specification.

Aru's ethical problem is not only that participant data could be misused. It is that participant expression could be transformed into unjustified authority.

The charter therefore governs both data and meaning.

25. Charter Principles

25.1 Epistemic Integrity

Aru must preserve the integrity of the path from utterance to public reflection.

No derived object may silently claim more than its source supports.

25.2 Object Separation

Aru must preserve the distinction between utterance, observation, interpretation, cluster, theme, synthesis, and public reflection.

Object types must not collapse.

25.3 Provenance and Epistemic Debt

Every derived object must carry its epistemic debt.

The system must preserve enough non-identifying provenance to audit transformations without retaining reconstructive participant material.

25.4 Non-Direction

Aru must not tell a community what to do.

It must not recommend, instruct, rank, nudge, diagnose, or generate action plans.

25.5 Non-Representativeness

Aru must not claim to represent the whole community.

The output is drawn from offered material. It is not a vote, poll, mandate, consensus, or statistically valid claim.

25.6 Local Containment

Raw participant material should remain within the local deployment environment.

External processing of raw material is prohibited unless an extraordinary constitutional process approves a specific exception.

25.7 Nightly Burn

Raw and reconstructive participant material must be destroyed through nightly burn.

The system may preserve only constitutionally approved non-identifying derived material and epistemic provenance.

25.8 Anti-Surveillance

Aru must not be used to monitor individuals, identify dissenters, map factions, detect ideological alignment, assess behavioural risk, or provide organiser intelligence.

25.9 Anti-Metric Capture

Aru must not display counts, percentages, rankings, trend scores, sentiment scores, heat maps, dashboards, or "top concerns."

25.10 Anti-Factionalisation

Aru must not harden fluid tensions into fixed factions.

It may surface differences in experience, but it must avoid labelling groups in ways that create social battle lines.

25.11 Contextual Discipline

Context may resolve meaning; it may not supply position.

Aru may use governed context to understand references, but not to infer participant belief from community priors.

25.12 Stewardship Over Ownership

Aru must be stewarded, not owned.

No founder, host, funder, technical operator, or institution should have unilateral authority to redefine the system's purpose.

25.13 Refusal

Aru must be able to refuse deployment.

It should refuse contexts where constitutional compliance, participant protection, local containment, non-direction, or epistemic integrity cannot be maintained.

Part IX - Constitutional Requirements

26. Purpose of the Constitution

Aru's constitution must govern interpretive authority.

The constitution is not only a privacy policy. It is not only a misuse policy. It is a control structure for the creation, transformation, retention, and publication of meaning.

It must answer:

What may the system collect?

What may it infer?

What object types may it create?

What may each object claim?

What must be burned?

What provenance may survive?

Who may review outputs?

Who may interpret the constitution?

What powers are separated?

What uses are prohibited?

When must deployment stop?

27. Constitutional Articles Required

A full constitution should include at least the following articles.

Article I - Purpose

Aru exists to preserve epistemic integrity across the transformation of anonymous participant expression into non-directive public reflection.

It does not exist to govern, direct, diagnose, survey, monitor, persuade, rank, predict, or manage a community.

Article II - Object Status

The constitution must define all primary object types and prohibit ontological collapse between them.

No utterance may become fact by transformation.

No cluster may become consensus.

No theme may become community belief.

No synthesis may become mandate.

Article III - Transformation Warrants

Every movement from one object type to another must be authorised by a transformation warrant.

Unwarranted transformations are invalid.

Article IV - Provenance

Every derived object must preserve non-identifying epistemic provenance sufficient for audit.

Content provenance capable of reconstructing participant utterances must burn unless explicitly and constitutionally exempted.

Article V - Local Operation

Raw participant material must remain local by default.

External processing is prohibited except through extraordinary constitutional review.

Article VI - Nightly Burn

Raw and reconstructive participant material must burn nightly.

Failure of the burn triggers review. Repeated or unexplained failure triggers suspension.

Article VII - Non-Direction

Aru must not produce recommendations, action plans, advice, behavioural nudges, mandates, or institutional instructions.

Article VIII - Anti-Metric Capture

Aru must not publish counts, percentages, rankings, trends, scores, or dashboards.

Article IX - Contextual Competence

Aru may use governed context to resolve meaning.

It may not use prior context to supply participant position.

Community-belief priors are prohibited.

Article X - Separation of Powers

The constitution must separate powers across capture, redaction, interpretation, aggregation, constitutional review, publication, audit, stewardship, and amendment.

Article XI - Stewardship

Aru must be stewarded under duties of restraint, participant protection, epistemic integrity, and constitutional fidelity.

Article XII - Constitutional Interpretation

Future interpretation of the constitution must favour non-direction, participant protection, epistemic humility, anti-retention, anti-capture, and limits on authority.

Article XIII - Refusal, Suspension, and Dissolution

Aru must refuse or suspend deployment when its constraints cannot be maintained.

If Aru abandons non-direction, local containment, nightly burn, anti-surveillance, object-status discipline, or epistemic integrity, it must cease operating under the name Aru.

Part X - Separation of Powers

28. Why Separation of Powers Is Necessary

Aru requires constitutional separation of powers because interpretive systems can be captured internally.

If the same actor controls prompt design, redaction rules, interpretation logic, aggregation criteria, output drafting, publication, and audit, the system can be steered while appearing compliant.

Separation of powers prevents a single actor from controlling the full path from participant expression to public meaning.

29. Powers to Separate

29.1 Capture Power

Controls how participant material enters the system.

Risk: steering through prompts, interface, booth placement, or consent framing.

29.2 Redaction Power

Controls what material is removed or masked.

Risk: deleting inconvenient meaning under the cover of safety.

29.3 Interpretation Power

Controls how observations become candidate meanings.

Risk: smuggling assumptions into interpretation.

29.4 Aggregation Power

Controls how material is clustered and synthesised.

Risk: converting repetition into importance or tension into conclusion.

29.5 Publication Power

Controls what the community sees.

Risk: making the output host-serving, critic-serving, or overly authoritative.

29.6 Audit Power

Controls review of the system's integrity.

Risk: audit becoming symbolic rather than real.

29.7 Amendment Power

Controls changes to the constitution.

Risk: convenience rewriting the constraints.

No single person or institution should hold all powers.

Part XI - Constitutional Interpretation

30. Interpretation Rules

Aru's constitution will require interpretation. Future stewards will face ambiguous cases.

The following interpretive rules should apply.

When in doubt, interpret in favour of non-direction.

When in doubt, interpret in favour of participant protection.

When in doubt, interpret against raw data retention.

When in doubt, interpret against external processing.

When in doubt, interpret against epistemic inflation.

When in doubt, interpret against host convenience.

When in doubt, interpret against claims of representativeness.

When in doubt, preserve uncertainty.

Founder intent may inform interpretation, but must not override constitutional purpose.

Operational usefulness is not sufficient justification for expanded authority.

Part XII - Failure Modes

31. Primary Failure Modes

31.1 Epistemic Inflation

Derived objects gain more authority than their sources justify.

31.2 Ontological Collapse

Different object types collapse into one another.

31.3 Provenance Collapse

The system can no longer explain how a public reflection was derived.

31.4 Directive Creep

Reflection slowly becomes advice, recommendation, or governance input.

31.5 Metric Capture

Counts, rankings, or sentiment scores become targets.

31.6 Prior-Driven Interpretation

Contextual knowledge begins to supply participant position.

31.7 Community-Belief Modelling

The system develops or applies assumptions about what the community usually thinks.

31.8 Institutional Capture

The host uses Aru to claim legitimacy or manage perception.

31.9 Factionalisation

The system hardens fluid differences into named groups.

31.10 Deletion Failure

Raw or reconstructive material survives beyond the nightly burn.

31.11 Audit Failure

The system burns raw material but retains too little epistemic provenance to verify integrity.

31.12 Poetic Overreach

The final reflection sounds more profound or certain than the evidence supports.

Part XIII - Deployment Model

32. Deployment Conditions

Aru should deploy only where:

- the host understands non-direction;
- participants can understand the consent boundary;
- local operation can be maintained;
- nightly burn can be verified;
- object-status discipline can be implemented;
- regional context can be governed;
- community-belief priors can be prohibited;
- public outputs can be protected from host alteration;
- stewards can refuse publication if necessary;

the community is not likely to be harmed by mediated reflection.

Deployment should be refused where:

- the host wants metrics;
- the host wants recommendations;
- the host wants raw data access;
- the host wants proof of having listened;
- the system cannot remain local;
- the nightly burn cannot be enforced;
- participant misunderstanding is likely;
- the output will probably be used as a mandate;

the community is too exposed, fragile, or institutionally captured.

33. Prototype Priorities

The first prototype should not aim to produce impressive public reflections.

It should prove the following:

Can participant material be captured without identity?

Can transcription and redaction operate locally?

Can object types be marked and preserved?

Can transformation warrants be implemented?

Can contextual information be used without supplying position?

Can raw and reconstructive material burn nightly?

Can non-identifying epistemic provenance survive?

Can the system generate bounded reflections without metrics, recommendations, or false representativeness?

Can human stewards audit the chain without accessing raw participant speech after burn?

If the prototype cannot prove these, it is not yet Aru. It is only an AI reflection experiment.

Part XIV - Reviewer Objections

34. Objection: This Still Lauanders AI Authority

This objection is serious.

Aru produces public language from participant material. That language may sound authoritative because it is coherent, synthetic, and machine-assisted.

The answer is not to deny the risk. The answer is to make the system's limits structural.

Aru must preserve object status, uncertainty, provenance, non-representativeness, and constitutional review. The public reflection must not sound like a verdict.

If Aru cannot prevent AI fluency from becoming false authority, the objection succeeds.

35. Objection: Auditability Conflicts With Deletion

Correct.

This is one of the central design tensions.

Aru must burn raw and reconstructive material while preserving enough epistemic provenance to audit transformations. The system must solve this through non-identifying provenance records, object-status metadata, transformation logs, and deletion verification that do not reconstruct participant speech.

If this cannot be done, Aru's architecture is incomplete.

36. Objection: Regional Context Will Become Stereotyping

This is a real risk.

Aru needs context to interpret local meaning. But context can become prior-driven interpretation.

The governing rule is therefore strict:

Context may resolve meaning; it may not supply position.

If Aru cannot enforce that rule technically and constitutionally, contextual adaptation should be limited or refused.

37. Objection: A Non-Directive System Still Influences Behaviour

Yes.

Any public reflection may influence behaviour. Non-direction does not mean zero influence. It means the system does not intentionally recommend, rank, instruct, or optimise behaviour.

Aru cannot guarantee that reflection has no effect. It can only ensure that it does not convert participant expression into direction.

The distinction is necessary but not complete. It should remain under review.

38. Objection: This Is Overbuilt

Possibly.

A simpler system would be easier to build. But a simpler system may not preserve the distinctions that make Aru legitimate.

If the project is merely a creative reflection booth, much of this architecture is unnecessary. But if the project is an epistemic integrity engine for civic reflection, the constraints are not decoration. They are the system.

The project should not use the stronger claim unless it is willing to build to that standard.

Part XV - Working One-Page Summary

39. Aru in Brief

Aru is a local epistemic integrity engine for non-directive civic reflection.

It receives anonymous participant expression and transforms it into public reflection under strict constitutional constraints.

Aru is not designed to discover what a community thinks. It is designed to constrain what may responsibly be said from the material participants offered.

Its central problem is epistemic integrity across transformation. Participant utterances become redacted utterances, extracted observations, candidate interpretations, clusters, themes, syntheses, and public reflections. Each stage creates risk. The system must prevent interpretation from becoming fact, repetition from becoming importance, synthesis from becoming consensus, and reflection from becoming mandate.

Aru operates locally by default. Raw participant material does not leave the deployment environment. Raw and reconstructive material burns nightly. Only constitutionally approved non-identifying derived material and epistemic provenance may survive.

Aru is non-directive. It does not recommend, rank, diagnose, advise, govern, monitor, or produce action plans.

Aru's key doctrine for contextual knowledge is:

Context may resolve meaning; it may not supply position.

Aru may use local context to understand references, idioms, and deployment-specific language. It must not use prior knowledge to infer what a participant believes.

Aru's legitimacy depends on object-status discipline, transformation warrants, provenance, uncertainty preservation, local containment, nightly burn, anti-surveillance, anti-metric capture, anti-factionalisation, and constitutional separation of powers.

The core claim is:

Aru exists to preserve the integrity of the path from utterance to public reflection.

Part XVI - Current Build Questions

40. Current Build Questions

The next stage of the paper should answer the following in more detail:

How exactly should object-status metadata be represented?

What minimum epistemic provenance is needed for audit?

How can provenance survive without becoming reconstructive?

What should the first prototype implement manually versus technically?

How should the constitutional review interface work?

What contextual knowledge may be loaded before deployment?

Who approves governed regional context?

How are transformation warrants generated, stored, and audited?

What is the minimum viable public reflection format?

What does a failed output look like?

What conditions require suspension?

These are not peripheral implementation questions. They define whether Aru can be built in accordance with its own frame.